ENGINEERING PORTFOLIO · 2026

Ilyes Laraqui Houssaini

Mechanical engineer · Design, data & modelling

I'm a mechanical engineer who enjoys a support role: using my skills to back up other people's work and help the team succeed. I've designed wind tunnel hardware in Formula 1 and led a Formula Student car as Technical Director. I'm now building the mathematical and programming side with an MSc in Financial Engineering at Imperial College London.

First
Class Honours, BEng Mechanical Engineering, UCL

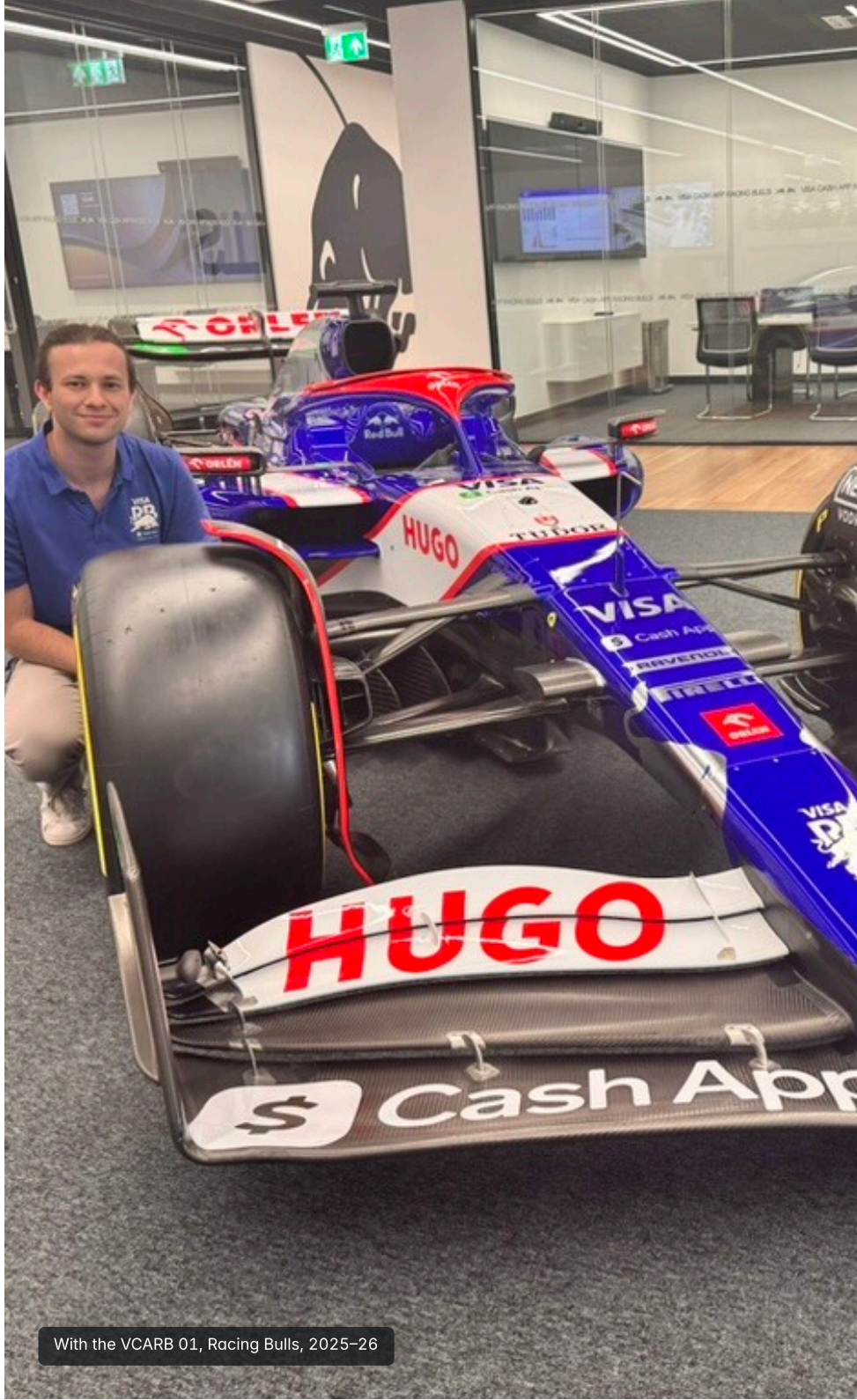
0.8 mm
FEA vs laser scan on F1 wind tunnel wheel parts

−26%
mass from my Formula Student front suspension

5.8 s
lap time gained by coaching with ECU data

01 Profile
02 Racing Bulls F1
03 Front suspension
04 Performance & data
05 Design & build
06 Modelling & code
07 Beyond engineering

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With the VCARB 01, Racing Bulls, 2025–26

EXPERIENCE

Mechanical Design Placement, Wind Tunnel Engineering · Racing Bulls F1
Jul 2025 – Jul 2026 · Year in Industry
Designed wind tunnel model hardware and test rigs in Siemens NX, with FEA and BS 8888 GD&T, then followed parts through manufacture, assembly and testing. Supported the race engineering team in the ops room on race weekends.

Technical Director · UCL Formula Student
Jul 2024 – Jul 2025
Led 10+ engineers across chassis, suspension, powertrain and engine mapping. Planned the build with a Gantt chart, managed resources and ran the cash-flow analysis.

Performance Leader & Design Engineer · UCL Formula Student
Feb 2023 – Jul 2024
Built telemetry and lap-time simulation tools, modelled the cooling system, designed the rear suspension and integrated sensors.

Production Methods Intern · Sweetch Energy, France
Aug 2024
Wrote a Python model of pump characteristics, redesigned a workshop layout and set up quality-control stations.

Mechanic Intern · BS Auto sports car garage, France
Dec 2018
Rebuilt a Porsche 911 engine at 14 and analysed a BMW gearbox. That is where my interest in engineering started.

EDUCATION

MSc Risk Management & Financial Engineering
Imperial College London · 2026 – present
Stochastic calculus, econometrics and financial statistics, derivatives pricing, risk management, computational finance in C++.

BEng Mechanical Engineering with Year in Industry · First Class Honours
University College London · 2022 – 2026
Highest marks: Advanced Thermodynamics & Fluid Mechanics (81.5), Manufacturing & Design (80.4), Individual Project (77.0), Application Programming for Data Science (75.8).

French Baccalaureate, top 5%
Lycée Français du Caire

SKILLS

Design	Siemens NX, Autodesk Fusion 360, BS 8888 GD&T, prototyping (laser cutting, 3D printing, bonding)
Analysis	NX FEA, Ansys, Abaqus, hand calculations, suspension kinematics
Modelling	MATLAB, Simulink / Simscape, IPG CarMaker
Code & data	Python (pandas, matplotlib, FastF1), SQL, C++ (MSc)
Quantitative	Stochastic calculus, statistics, Monte Carlo simulation
Languages	French (native), English (professional), German (intermediate)



UCL Racing 2024 car. I was Performance Leader that year and Technical Director the next.

HOW I WORK

The same loop I've used on parts, rigs, models and processes:

1 Define the problem	**2** Understand the status quo	**3** Design options	**4** Agree a direction with the team	**5** Simulate and test	**6** Correlate with real data	**7** Peer review	**8** Implement at the right moment	**9** Monitor, then close

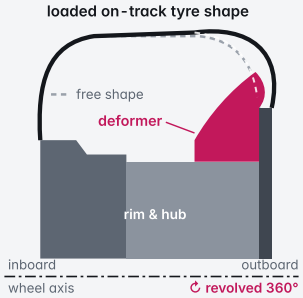
02 Racing Bulls F1: Wind Tunnel Mechanical Design

A year in the Wind Tunnel Engineering group, designing the hardware the aero team uses to test new ideas, from model parts to test rigs.

Year in Industry · Jul 2025 – Jul 2026
Siemens NX · NX FEA · BS 8888 GD&T

Wheel assembly & tyre deformers

Tyre shape affects diffuser and rear brake duct flow, so the tunnel tyre must match its on-track shape. From the aero surfaces, I designed the rims, hubs, wheel covers and deformers that push the outboard shoulder out to that shape (one profile, revolved 360°). I owned them end-to-end: release, assembly and balancing, and fitting to the model.

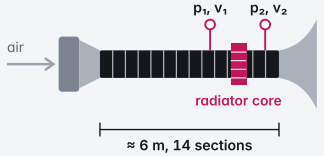


My own illustration, not the actual part

0.8 mm max shape difference, FEA vs laser scan

Radiator characterisation rig

Model radiators are tested on a rig and correlated with CFD. I designed a full-size version of that rig, a 6 m "mini wind tunnel" of 14 bonded sections holding a real radiator core, to measure **pressure and velocity loss across the core** and correlate the model radiator with the real one.



correlate model core ↔ real core

My own illustration, not the actual rig

6 m rig, 14 bonded sections, full-size radiator core

A 5-year assembly problem, fixed

One model assembly had been unreliable for more than five years. I traced the cause to clearances tighter than **0.010 mm** that no drawing captured. I added the correct GD&T annotation to the drawing and checked the fix over 15 assembly iterations.

15 / 15 assemblies correct after the drawing was fixed

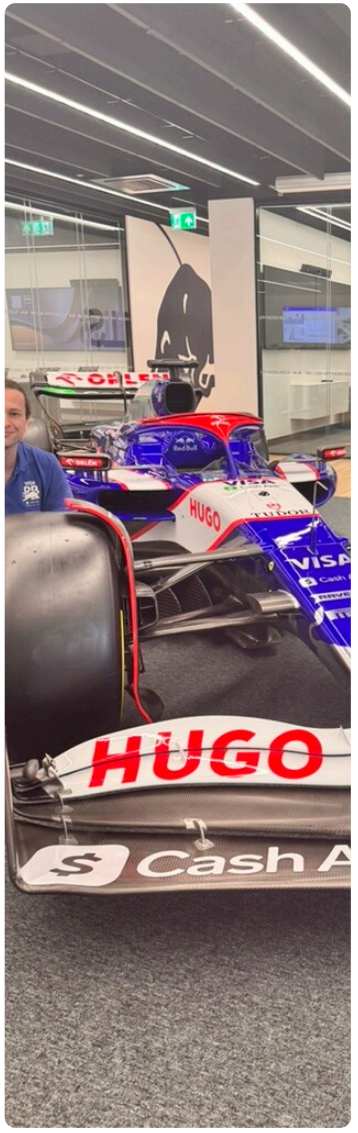
Overnight fix for a part clash

At 20:00 I was the last mechanical engineer on site when the Model Shop found a part clash that had to be fixed before a 06:00 tunnel run. I up-issued the part, issued the updated drawing within 3 hours, and stayed until it was manufactured and fitted to the model for checking. This allowed the test of an aero component that then went on a **quick release as a track upgrade**.

3 h from clash found to updated drawing issued

Test rigs, model parts and race weekends

- Designed several rigs for testing new technologies for the tunnel model, including a calibration rig and bench-test rigs.
- Designed suspension pick-up parts for the model, checked against kinematic points and for clashes.
- Improved the model's metrology datum system for faster, more convenient alignment and tracking.
- On race weekends, supported the race engineering team in the ops room by monitoring competitor radio.



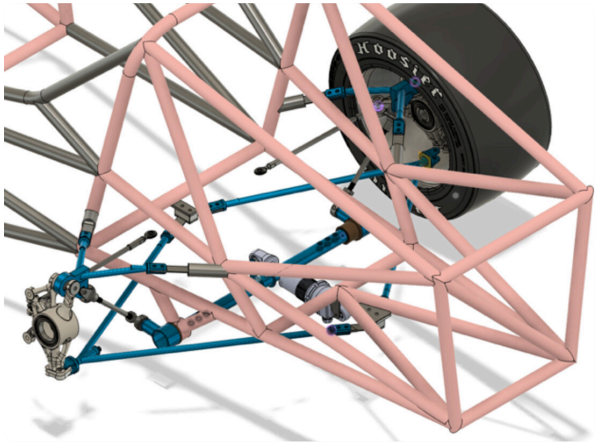
At Racing Bulls, 2025–26

NDA No Racing Bulls hardware images are shown. The diagrams are my own simplified illustrations.

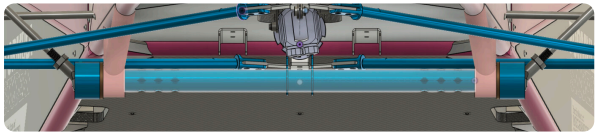
03 Front suspension: pull-rod mono-shock

My individual dissertation, designed for the UCL Formula Student car while I was Technical Director.
Mark: 77/100.

UCL Formula Student · 2024–25
Fusion 360 · MATLAB · kinematic simulation · hand calculations



Pull-rod front corner, re-using the existing uprights



One damper for both wheels; the rocker works as an adjustable torsion bar



Kinematic prototype I built



Blade flexures on the car

The problem. UCL's front suspension hadn't changed since 2014: push-rod double wishbones with a damper per corner. It was heavy and costly, and drivers reported low grip in fast corners.

- **Geometry from camber gain.** I doubled the target to −0.8°/in (from −0.4°/in), set the pick-up points from the swing-axle length (1819 mm) and checked them in kinematic simulation.

  camber gain **−0.78°/in** · motion ratio **1.42** · roll centre **42 mm** · anti-dive **11%** · bump steer **0.05 mm/in**

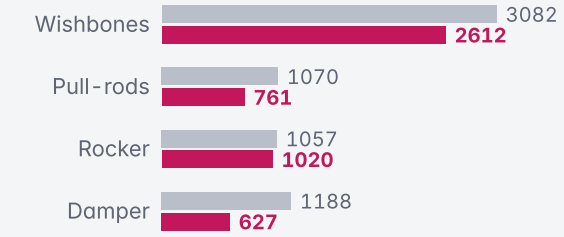
- **Mono-shock pull-rod.** One damper for both wheels lowers the centre of mass. The shared rocker acts as a torsion bar in roll, so roll stiffness can be tuned separately from heave by choosing one of three pull-rod mounting holes.
- **Sized for the worst loads.** I hand-calculated braking, acceleration, cornering and 5 g kerb-strike loads (members up to 11.7 kN), then chose T45 wishbones, 4130 tie rods and EN24T steel.
- **Flexures instead of spherical bearings.** I sized blade flexures in a MATLAB script using large-deflection beam theory and checked bending, von Mises and buckling safety factors. Flexures have no stiction and save mass and cost.
- **Checked on a physical prototype.** I built a plywood and 3D-printed rig to measure the kinematics. Camber gain matched the simulation to within 0.22° across the full wheel travel.

Target	Result
Cut system mass by 10%	**−26.2%** (7.73 → 5.70 kg)
Cut system cost by 10%	**−9.2%** (£6,724 → £6,108)
Fewer parts, simpler manufacture	Fewer rockers, moving parts and spherical bearings
Lower centre of mass · tunability · modular	Pull-rod layout · torsion-bar rocker · mounts to car front

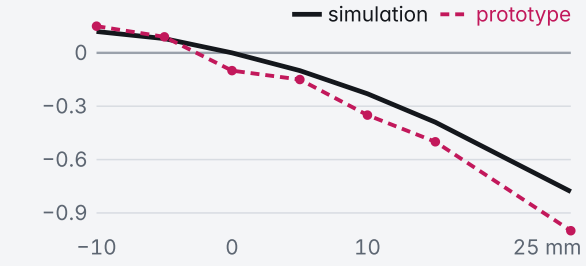
−26.2%
system mass
(target −10%)

−9.2%
system cost

Mass by assembly (g): old design vs mine



Camber change (°) vs wheel travel

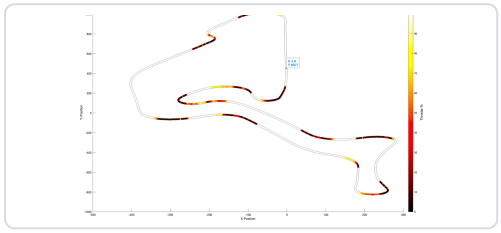


04 Vehicle performance: data, models and simulation

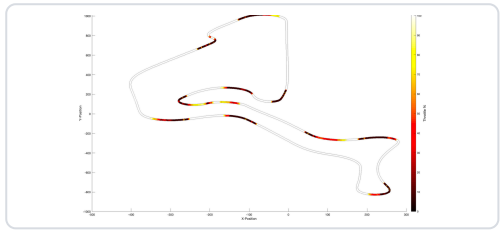
Turning logged data and models into decisions: where drivers lose time, why the car overheats, which setup scores best.

Performance Leader, UCL Formula Student · 2023–25
MATLAB · Simscape · IPG CarMaker

Telemetry to track maps, then driver coaching



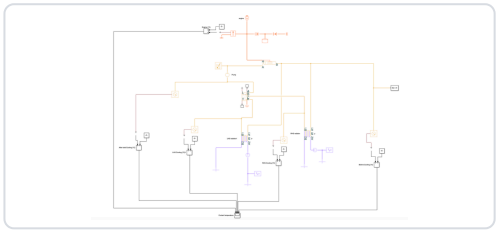
Throttle map, generic setup (Sim Series data, first check of the tool)



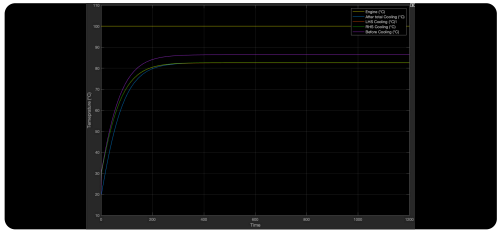
Throttle map, updated setup

I wrote MATLAB scripts that turn logged throttle, brake, gear, ride height and roll/yaw rates into track heat maps. I first checked the tool on FS Sim Series data, then ran it on the car's LifeRacing ECU data, overlaying two drivers to show where time was lost and where to brake and upshift. The slower lap came down from **1:10.623 to 1:04.832**.

Cooling model and sidepod redesign



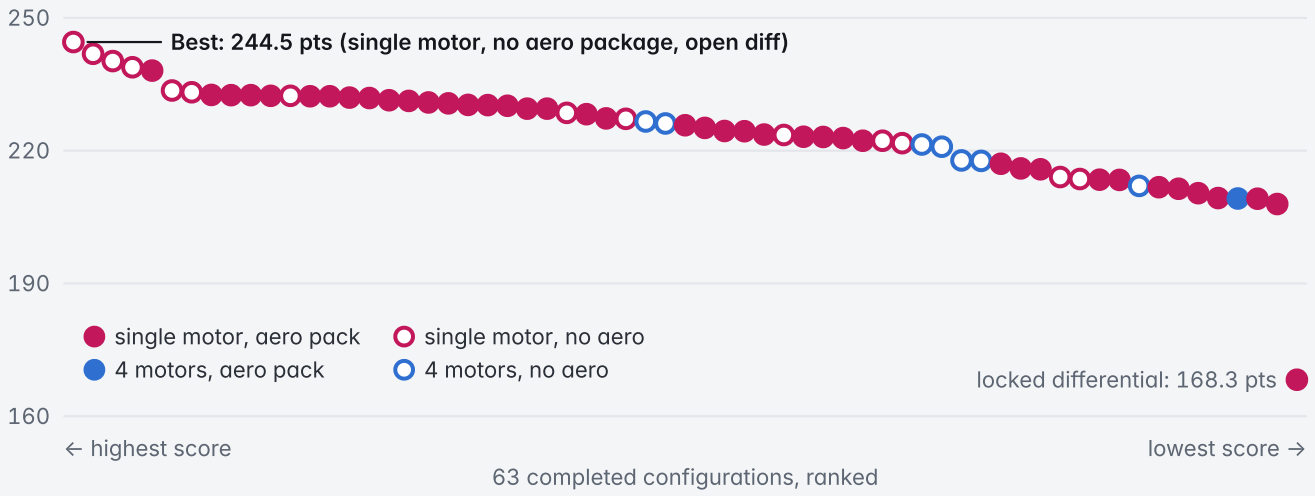
Simscape model of the coolant loop



Predicted temperatures with the full core in use: 80–85 °C

Coolant passed 100 °C after 5 endurance laps and peaked at 120 °C. My Simscape model showed only about **30% of the radiator core** was getting useful airflow. That led to a sidepod redesign sending all the air through the core, and coolant came down to **85 °C** at 20 °C ambient.

Lap-time simulation: combined event score of each configuration (points)



Lap-time simulation as a design of experiments

For the FSUK lap-time simulation event, I set up **78 vehicle configurations** in IPG CarMaker. They varied powertrain, aero package, differential, suspension geometry, toe and camber. I scored each one on Acceleration, Skidpad and Sprint, then ranked the 63 that completed.

The ranking also showed which choices mattered most. The single-motor layout without the aero package scored highest, and toe and camber separated the top runs.

23rd / 57 our result in the following year's lap-time simulation event

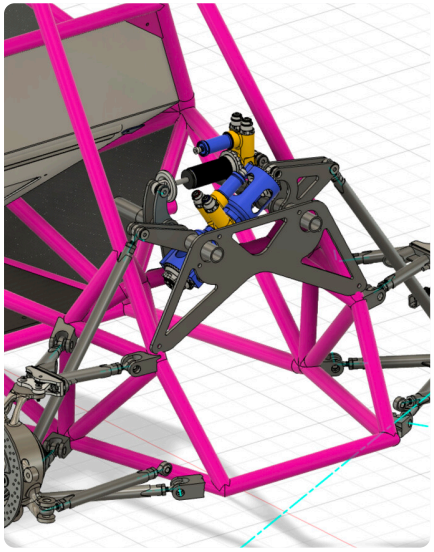
05 Design and build at UCL Formula Student

Parts that had to be made by students, fit a crowded chassis, and survive a race weekend.

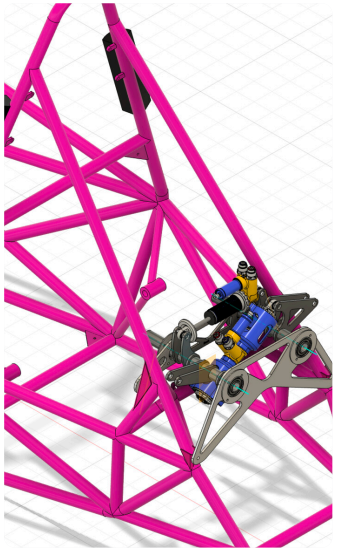
Design Engineer, UCL Formula Student · 2022–24
Fusion 360 · laser cutting · 3D printing

Rear suspension with separate heave and roll

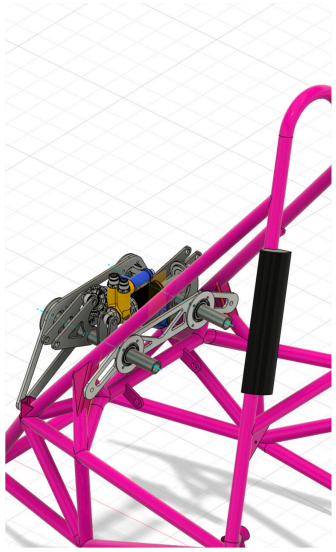
I designed a rear push-rod layout around Multimatic heave and roll dampers. Separating the two means heave stiffness (which controls pitch and ride height) can be tuned without changing roll stiffness. I built a moving kinematic model in Fusion 360 to check the rocker motion and clearances in pure heave and pure roll before anything was made.



Full rear assembly with uprights and brakes



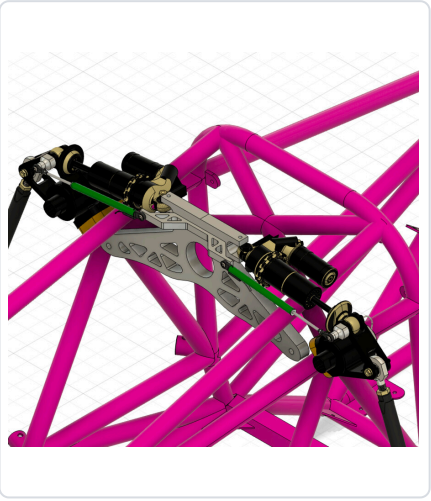
Heave/roll rocker package



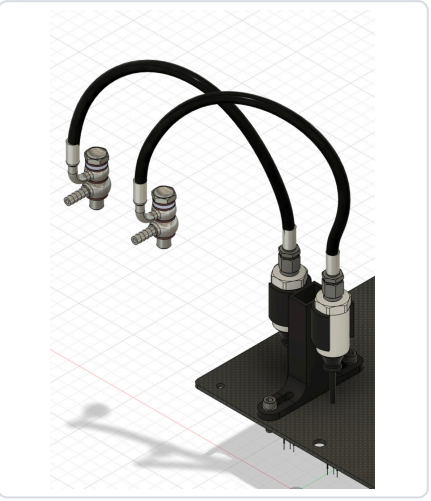
Packaging inside the frame

Earlier projects

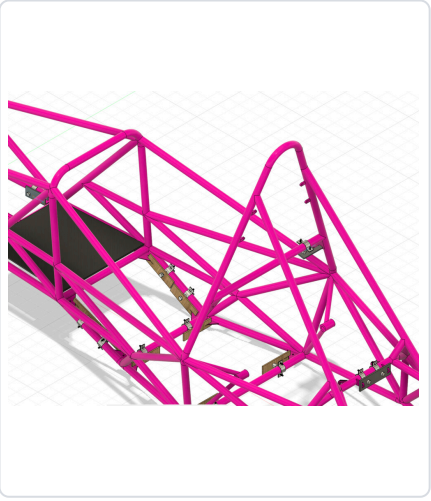
- **IMechE Design Challenge (national):** designed a pipe-climbing robot in Fusion 360 under a £50 cost cap, estimated climb time in Excel, and built it with drills, laser cutters and 3D printers.
- **UCL Mechathon (48 h):** designed and made the counterweight and hook for a hydraulic arm.



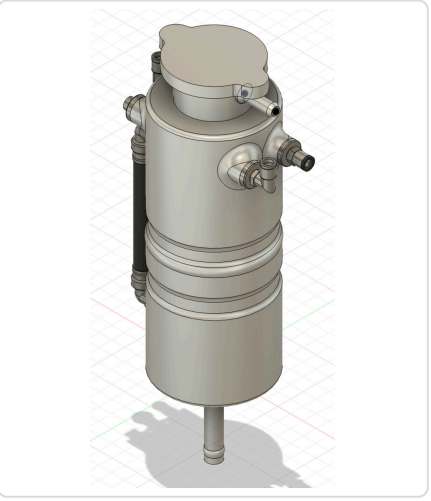
Suspension travel sensor mounts for logging damper travel



Brake pressure sensor mounts



Chassis tab jigs to hold tabs in place for welding



Cooling tank, reverse-engineered and modelled in CAD

06 Modelling, code and quantitative work

The analytical side behind the hardware: building models, checking them against reality, and writing the code to do it.

UCL · Imperial · Sweetch Energy
MATLAB · Python · Abaqus · C++

IMPERIAL COLLEGE LONDON

MSc Risk Management & Financial Engineering

Stochastic calculus, econometrics, derivatives pricing and risk management, with Monte Carlo simulation in C++. The common thread with engineering is making decisions from noisy data under uncertainty, whether the question is a price, a risk or a race strategy.

stochastic calculus · Monte Carlo · C++ · econometrics

UCL · DYNAMICS & CONTROL

Controller and vibration design in MATLAB

Designed a PD controller that cut overshoot by about 21% and response time by about 80% compared with proportional control. Built a six-mass vibration model, solved it in MATLAB and designed mounts that hit a 10.2% response reduction at 23 Hz within mass and stiffness limits.

MATLAB · transfer functions · multi-DOF dynamics

UCL · ELASTICITY & PLASTICITY

Nonlinear FEA checked against experiment

Processed tensile-test data in MATLAB (E = 63.9 GPa, yield 116 MPa) and built an elastic-plastic-damage model in Abaqus with a mesh-convergence study. My predicted 54.7° shear-fracture angle matched both the FEA and lab footage of the real specimen.

Abaqus · MATLAB · validation

SWEETCH ENERGY · INTERNSHIP

Pump model for production methods

Wrote a Python model of pump characteristics. I also redesigned a workshop layout and set up quality-control stations.

Python · process design · quality control

UCL · DATA SCIENCE MODULES

Cleaning data and building small apps

Cleaned and explored a London air-quality dataset with pandas (mixed types, missing values, mis-parsed percentages), designed an SQLite database for it, and built simple Dash and Flask apps on top.

pandas · SQL · Dash / Flask

PERSONAL PROJECT

Exploring F1 timing data

In my own time I use the FastF1 library to pull public F1 telemetry: overlaying speed traces of the same driver's qualifying laps across seasons, and mapping gear use around a lap to see where cars differ.

Python · FastF1 · matplotlib

07 Leadership, people and the rest of me

What I do around the work: leading a team, helping others learn, and the things that keep me curious.



The UCL Racing team with our car at Silverstone

Leading a student race team

As Technical Director I led 10+ engineers across chassis, suspension, powertrain and engine mapping. I set the design priorities around mass, cost and tunability, planned the build and resources with a Gantt chart, and ran the cash-flow analysis.

Helping others

- UCL Mechspace mentor, helping students on the workshop machines
- F1 in Schools judge and FIA Girls on Track volunteer (2024–26)
- UCL transition mentor for first-year mechanical engineers (2023)
- French literature tutor, and organiser of a food and money collection for refugees, Cairo (2020–21)

About me

I'm French and Moroccan and grew up moving every four years or so: London, Fes, Zurich, Cairo, then back to London. Changing countries that often taught me to settle into new teams quickly and to work across languages and cultures.

My interest in engineering started in a sports car garage in Alsace, where I rebuilt a Porsche 911 engine at 14.

Outside work

Motorsport: race strategy and car physics · Rubik's cube (3×3 best: 42 s) · Football, basketball, handball · Karate, blue belt · Coding and algorithms

Languages

French (native) · English (professional) · German (intermediate)

GET IN TOUCH

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